COMSATS University Islamabad

Abbottabad Campus

Graded LAB Task

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Class:

BCS-4B

Registration:

FA24-BCS-112

Subject:

AI(Artificial Intelligence)

Instructor

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- **Lab Task 1**: Student Grade Calculator. Write a Python program to accept marks of a student **(1-100)** and display the grade according to the following criteria:

- Grade F if marks are less than **50**
- Grade E if marks are between **50 to 60**
- Grade D if marks are between **61 to 70**
- Grade C if marks are between **71 to 80**
- Grade B if marks are between **81 to 90**
- Grade A if marks are between **91 to 100**

CODE:

```
marks = int(input("Enter your marks (1-100): "))
if marks < 50:
    print("Grade: F")
else:
    if marks <= 60:
        print("Grade: E")
    else:
        if marks <= 70:
            print("Grade: D")
        else:
            if marks <= 80:
                print("Grade: C")
            else:
                if marks <= 90:
                    print("Grade: B")
                else:
                    if marks <= 100:
                        print("Grade: A")
                    else:
                        print("Invalid marks")
```

OUTPUT:

```
PS C:\Users\dell\OneDrive\Hình ảnh> & C:
Downloads/Untitled-1.py
Enter your marks (1-100): 44
Grade: F
PS C:\Users\dell\OneDrive\Hình ảnh> \
```

```
PS C:\Users\dell\OneDrive\Hình ảnh>
Downloads/Untitled-1.py
Enter your marks (1-100): 98
Grade: A
PS C:\Users\dell\OneDrive\Hình ảnh>
```

- **Lab Task 2**: Factorial Calculator .Write a program that takes a number from the user and calculates the factorial of that number:

CODE:

Untitled-1.py

C: > Users > dell > Downloads > Untitled-1.py > ...

```python
num = int(input("Enter a number: "))

fact = 1

for i in range(1, num + 1):
    fact = fact * i

print("Factorial is:", fact)
```

OUTPUTS:

```
PS C:\Users\dell\OneDrive\Hình ảnh> & C:/Users/dell/
Downloads/Untitled-1.py
Enter a number: 12
Factorial is: 479001600
PS C:\Users\dell\OneDrive\Hình ảnh>
```

```
PS C:\Users\dell\OneDrive\Hình ảnh> & C:/Users/dell/Ap
Downloads/Untitled-1.py
Enter a number: 5
Factorial is: 120
PS C:\Users\dell\OneDrive\Hình ảnh>
```

- **Lab Task 3:** Fibonacci Series Fibonacci series is defined as a sequence where the next number is formed by adding the previous two numbers. Start from **0** and **1.**

Example**: 0, 1,1,2,3,5,8,13,21,34, 55**

Steps:

- Take an input number that shows how many terms to display.
- 2. Use loops to display the Fibonacci series up to that term.
- 3.Example: If the input number is 6, the output should be**: 0,1,1,1,2,3,5**

CODE:

```
Untitled-1.py
C: > Users > dell > Downloads > Untitled-1.py > ...
n = int(input("Enter Numbers For Which you want to make Series: "))

a = 0
b = 1
count = 0

while count < n:
    print(a)

    c = a + b
    a = b
    b = c

    count = count + 1
```

OUTPUTS:

```
PS C:\Users\dell\OneDrive\Hình ảnh> & C:/Users/dell/AppData/Local/Programs/Pyt
Downloads/Untitled-1.py
Enter Numbers For Which you want to make Series: 4
0
1
1
2
PS C:\Users\dell\OneDrive\Hình ảnh>
```

```
PS C:\Users\dell\OneDrive\Hình ảnh> & C:/Users/dell/AppData/Local/Pr
Downloads/Untitled-1.py
Enter Numbers For Which you want to make Series: 6
0
1
1
2
3
5
PS C:\Users\dell\OneDrive\Hình ảnh>
```

The End

GitHub link: https://github.com/users/uk808382-dev/projects/1